DA08 Select, filter and calculate a revenue column

Week 3, Tables in pandas | Session 8 of 24 | Python for Data Analysis

Today's question. How much money did the practice shop take in, and where did it come from?

Objective. By the end: you can pick a column, keep the rows that meet a condition, and add a revenue column.

> WHERE THIS IS USED
> A shop owner asks which sales were big and how much money they brought in. Picking out rows and adding a new column is how analysts answer that on their own data. Which rows of a shop table would you pick out first?

Before this session: DA07 A table in Python: DataFrame, columns and reading a CSV. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
select	Name a column in square brackets to look only at it.	Reading only the price strip of the register.
filter	Keep only the rows where a comparison is True.	Picking out the sales of 5 units or more.
mask	A column of True and False values, one for each row, that a filter uses.	A ticked list beside the register.
new column	A column worked out from other columns and added to the table.	Multiplying units by price on every line.
revenue	Money received from sales: units times price. It is not profit.	The takings before paying for stock.
sum	All the values added together.	Adding up the day's takings.

Block A. Select and filter

Read the practice shop. Square brackets with a column name select one column. A comparison inside square brackets keeps only the rows where it is True.

Predict. Predict how many of the 60 sales have 5 units or more. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
shop = pd.read_csv("camp_shop.csv")
units = shop["units"]
big = shop[shop["units"] >= 5]
```

```
print(len(big))
print(units.sum())
```

Run it, then compare. The cell prints 30 and then 277. shop["units"] selects one column. The comparison gives True or False for every row, and only the True rows stay. All 60 sales together sold 277 units.

Investigate.

1. Is the filter deleting the other 30 rows?
2. Why can the sum of units not tell us the money earned?

Modify. Filter the Juice sales with 5 units or more and count them.

Block B. A new revenue column

Assigning to a new column name creates the column. Multiplying two columns multiplies their matching rows. This is how units times price becomes revenue.

Predict. Predict the total revenue, using the item prices 3000, 1000 and 2000. Write your guess on paper first.

Your guess, written before you run it

```
shop["revenue_ugx"] = shop["units"] * shop["unit_price_ugx"]
print(shop.head(3))
print(shop["revenue_ugx"].sum())
```

Run it, then compare. The first three revenues are 3000, 2000 and 8000. The total revenue is 550000 shillings. This is money taken in. There is no cost column, so profit cannot be worked out.

Investigate.

1. Why is revenue not profit?
2. Does the new column change camp_shop.csv?

Modify. Find the revenue of the Book sales only.

Make

Keep the Pen sales that have 4 units or more. Count them and add up the revenue they bring in. Say why the count and the revenue are different quantities.

Save your work as DA08. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
KeyError: 'unit'	The column name is spelled differently from the header.	Print list(shop.columns) and copy the exact name.
The filter returns an empty table	The comparison matched no rows, often because of a typing slip in a text value.	Check the spelling and capital letters of the value you compare with.

Exit ticket

A filter keeps the rows I ask for, and a new column adds a calculation to every row.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.